

Skills Summary

Approximating Square Roots

Use this reference while completing your worksheet or when studying.

Skill 1 — Bracketing a root between whole numbers

Rule: squaring keeps order. If $p^2 < N < q^2$ then $p < \sqrt{N} < q$.

Method: find the perfect square just below the radicand and the one just above it. Their roots are the two consecutive whole numbers the root lies between.

Squares to have ready: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.

Example: Between which whole numbers does $\sqrt{52}$ lie?

Step 1. The radicand is not a perfect square, so no exact value exists — the useful answer is the pair of squares it falls between.

Step 2. $49 < 52 < 64$, and taking square roots across the inequality keeps the order.

$$\Rightarrow 7 < \sqrt{52} < 8$$

Try it: Between which whole numbers does $\sqrt{85}$ lie?

check: between 9 and 10

Skill 2 — Squeezing a root to a decimal

Method: once the root is bracketed, test decimals inside the bracket by *squaring* them. When two neighbouring decimals trap the radicand, the root is between them — round to whichever is asked for.

Squaring a candidate is the check. Never round a root you have not squared back.

Example: Give $\sqrt{40}$ to the nearest hundredth.

Step 1. The radicand sits between two perfect squares, so I squeeze inside that bracket by squaring tenths, then hundredths.

Step 2. $6.3^2 = 39.69$ and $6.4^2 = 40.96$, so the root is in the low sixes; then $6.32^2 = 39.9424$ and $6.33^2 = 40.0689$ trap it.

$$\Rightarrow \sqrt{40} \approx 6.32$$

Try it: Give $\sqrt{55}$ to the nearest hundredth.

check: 7.42

(!) Watch out: halving the radicand is not a square root. Always square your estimate and compare it with the number under the radical — that one check catches almost every wrong answer on this topic.

Skill 3 — Using the estimate in a measurement

Rule: a square of area A has side $\sqrt{A}$, in the matching length unit — square feet give feet, square metres give metres.

Round only at the end, and write the unit on the answer. When material has to *cover* the area, round **up** to the next whole size.

Example: A square patio covers 90 square feet. Give its side to the nearest tenth of a foot.

Step 1. Area of a square is side times side, so the side is the square root of the area — and the root is irrational, so the answer is an estimate.

Step 2. $9^2 = 81$ and $10^2 = 100$ bracket it; squeezing with tenths, $9.4^2 = 88.36$ and $9.5^2 = 90.25$ trap the area.

$\Rightarrow$ side $\approx$ **9.5** ft

Try it: A square deck covers 130 square feet. Give its side to the nearest tenth of a foot.

check: 11.4 ft